

Manufacturing Engineering Technology I

TEKS §127.813 (working draft) · CTE Level 2 · Grade 10 · Mr. Gavaldon · Da Vinci School

Week 2
MONDAY Day 6 - Measurement & Tools

Aug 17

OBJECTIVE	Students will measure real objects with appropriate tools and explain precision, accuracy, and tolerance.
TEKS	§127.813 (draft, verify strand) measurement, tools & precision in manufacturing
ELPS / EB SUPPORT	Tool use is demonstrated + hands-on (VC). Station cards use pictures; groups discuss readings (PN, CD).
AGENDA · 45 MIN	5' Bell-work: how small can you measure by eye? 12' Calipers, speed square, precision vs tolerance 20' Measuring stations: measure & record real objects 8' Compare results + exit
SLIDES / ACTIVITY	Measure Like a Machinist
NOTEBOOK	Record your station measurements. Define precision vs. tolerance in your own words.
TURN IN	Measuring Lab - Data Sheet

TUESDAY Day 7 - Manufacturing Math

Aug 18

OBJECTIVE	Students will perform unit conversions and fraction/decimal calculations used to specify manufactured parts.
TEKS	§127.813 (draft, verify strand) applied mathematics & measurement in manufacturing
ELPS / EB SUPPORT	Worked examples first, then practice (I do / we do / you do). Conversion table provided to keep (CD, RS).
AGENDA · 45 MIN	5' Bell-work: convert 2.5 in to mm 12' Conversions, fractions/decimals, tolerance math 20' Practice set (real part dimensions) 8' Go over answers + exit
SLIDES / ACTIVITY	The Math Behind the Part
NOTEBOOK	Keep the conversion table. Work 2 example problems with all steps shown.
TURN IN	Manufacturing Math - Practice

WEDNESDAY Day 8 - Parametric Design in Fusion

Aug 19

OBJECTIVE	Students will define parameters in Fusion 360 and explain why parametric modeling matters for manufacturing.
TEKS	§127.813 (draft, verify strand) parametric CAD for manufacturable parts
ELPS / EB SUPPORT	Steps mirrored on-screen and in Classroom screenshots (VC). Finishers help peers (PN).
AGENDA · 45 MIN	5' Bell-work: why change one number and everything updates? 12' What 'parametric' means + watch Fusion intro 22' Guided: build a plate driven by parameters 6' Save + exit
SLIDES / ACTIVITY	One Number Changes Everything
NOTEBOOK	Write the 3 parameters you set (name = value) and what each controls on the part.
TURN IN	— (notebook / no turn-in)

THURSDAY**Day 9 - Start the Parametric Project (Mounting Plate)**

Aug 20

OBJECTIVE	Students will plan the dimensions of a mounting plate and begin modeling it parametrically in Fusion 360.		
TEKS	\$127.813 (draft, verify strand) apply parametric CAD to a defined part; documentation		
ELPS / EB SUPPORT	Part-planning sheet organizes dimensions before CAD (CD). Reference model shown on screen (VC).		
AGENDA • 45 MIN	5' Bell-work: name 2 parts held on by a plate	10' Project brief + dimensions	
	25' Plan dimensions -> begin modeling in Fusion	5' Save + exit	
SLIDES / ACTIVITY	Project: Mounting Plate		
NOTEBOOK	Record your plate dimensions and hole sizes. Sketch the plate with dimensions labeled.		
TURN IN	Part Planning Sheet - Mounting Plate		

FRIDAY**Day 10 - Continue Project + Certification Kickoff**

Aug 21

OBJECTIVE	Students will continue their parametric plate and describe the Autodesk certification pathway and its domains.		
TEKS	\$127.813 (draft, verify strand) industry certification pathway; collaboration		
ELPS / EB SUPPORT	Cert path shown as a simple roadmap graphic (VC). Peer helpers pair with EB students (PN).		
AGENDA • 45 MIN	5' Bell-work: what does 'certified' get you?	10' The Autodesk certification path (5 exam domains)	
	25' Keep modeling your plate + help a neighbor	5' Self-check + exit	
SLIDES / ACTIVITY	Where This Is Headed: Certified		
NOTEBOOK	List the certification domains. Note where you are on your plate and one thing to finish Monday.		
TURN IN	Certification Path - Self-Check		